

Challenge (Habanero)

- Q1.** In a gaming tournament, the score s is determined by the number of wins w and losses l , where $s \geq 2w - 3l$. Which region satisfies this inequality?
- (A) The region below the line $s = 2w - 3l$
(B) The region above the line $s = 2w - 3l$
(C) The region on the line $s = 2w - 3l$
(D) The region between the lines $s = 2w - 3l$ and $s = 3w - 2l$
(E) The region outside the lines $s = 2w - 3l$ and $s = 3w - 2l$
- Q2.** What type of boundary line is used for the inequality $x + 2y > 5$?
- (A) Solid
(B) Dashed
(C) Dotted
(D) Dash-Dot
(E) None of the above
- Q3.** Which linear inequality has a solid boundary line?
- (A) $x + y \leq 4$
(B) $x + y < 4$
(C) $x + y > 4$
(D) $x + y \geq 4$
(E) $x + y = 4$
- Q4.** In a data usage plan, the cost c is determined by the number of gigs g used, where $c \leq 2g + 10$. Which region satisfies this inequality?
- (A) The region above the line $c = 2g + 10$
(B) The region below the line $c = 2g + 10$
(C) The region on the line $c = 2g + 10$
(D) The region between the lines $c = 2g + 10$ and $c = 3g + 5$
(E) The region outside the lines $c = 2g + 10$ and $c = 3g + 5$
- Q5.** What is the solution region for the system of linear inequalities $x + y > 2$ and $x - y < 1$?
- (A) The region above the line $x + y = 2$ and below the line $x - y = 1$
(B) The region below the line $x + y = 2$ and above the line $x - y = 1$
(C) The region on the line $x + y = 2$ and on the line $x - y = 1$
(D) The region between the lines $x + y = 2$ and $x - y = 1$
(E) The region outside the lines $x + y = 2$ and $x - y = 1$
- Q6.** Which of the following linear inequalities has a solution region that is unbounded?
- (A) $x + y \leq 5$
(B) $x + y < 5$
(C) $x + y > 5$
(D) $x + y \geq 5$
(E) $x + y = 5$
- Q7.** In a coding logic problem, the number of steps s is determined by the number of loops l , where $s \geq 3l - 2$. Which region satisfies this inequality?
- (A) The region below the line $s = 3l - 2$
(B) The region above the line $s = 3l - 2$
(C) The region on the line $s = 3l - 2$
(D) The region between the lines $s = 3l - 2$ and $s = 2l - 1$
(E) The region outside the lines $s = 3l - 2$ and $s = 2l - 1$
- Q8.** What type of boundary line is used for the inequality $2x - 3y \leq 1$?
- (A) Solid
(B) Dashed
(C) Dotted
(D) Dash-Dot
(E) None of the above

Open Ended Questions (FRQ)

- Q9.** Graph the linear inequality $x + 2y > 5$ and identify the solution region. Show all work and explain your reasoning.
- Q10.** Solve the system of linear inequalities $x + y \leq 4$ and $x - y \geq -2$. Graph the solution region and identify the boundary lines. Show all work and explain your reasoning.

Chef's Tips

- Tip 1: Emphasize the importance of solid and dashed boundary lines in linear inequalities.
- Tip 2: Use real-world examples from tech and gaming to illustrate the application of linear inequalities.
- Tip 3: Encourage students to show all work and explain their reasoning when solving free-response questions.